Acting Locally, Leading Globally with Build Now: An interdisciplinary, collaborative research and training program focused on affordable housing innovation in Waterloo Region

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Executive Summary:

Background: Canada faces a growing housing affordability crisis. Average home prices have surged 77% in a decade, now exceeding $710,000—far outpacing median incomes and locking many younger Canadians out of ownership. This erosion of generational equity has even been flagged by the RCMP as a national security concern. The crisis is driven by multiple factors, including rapid population growth, limited land and housing supply, regulatory constraints, and rising construction costs exacerbated by labour shortages.

Despite the long-term benefits of home ownership—financial security, social mobility, health, and well-being—Canadian housing policy remains focused primarily on rentals. Meanwhile, housing development has major environmental consequences, from habitat loss and emissions during construction to long-term transportation impacts.

Pitch: Addressing the challenge of creating affordable, suitable, and sustainable home ownership in Canada requires a fundamental shift in how we think about and build housing. The Future Cities Institute proposes a world-class Living Lab in Waterloo Region to advance a transformational affordable housing model. This initiative leverages the *Build Now Waterloo Region* initiative, a $500-million initiative led by Habitat for Humanity to build 10,000 new "missing middle" homes—70% for ownership—by 2030.

Through interdisciplinary research, experiential learning, and local and international partnerships, the Build Now Living Lab will assess and amplify the economic, health, sustainability, technological and societal impacts of community-led housing — creating local solutions with global relevance.

Global Futures Funds will leverage at least $500,000 in external funding to support research, training, and partnership development. Research activities include the Global Futures Seed Funding competitions and FCI Postdoctoral research pilot funds. Training activities include future cities capstones, CIVIC co-op positions, and Build Now Innovation Challenges. Finally, service and partnership development activities include City Sprints, and Capacity Development through the European Network of Living Labs (ENoLL).
Currently, over 40 faculty members from all six Faculties are participating in our proposal. External partners include local (e.g., the Waterloo Region Community Foundation, the Build Now team), and international experts (e.g., ENoLL).

The Build Now Living Lab directly addresses a nationally urgent, socially complex challenge with the potential for real and lasting impact. It proposes a novel model of interdisciplinary, data-informed, and community-embedded research within a large-scale real-world development. It brings together academia, civil society, industry, and government in a true partnership, positioning our Region as a global leader in housing innovation. The Living Lab will provide the evidence, tools, and scalable models needed to fundamentally reshape how communities are built—transforming housing from a source of crisis to a platform for prosperity, well-being, and sustainability.

VISION: The Future Cities Institute envisions a world-class Living Lab in Waterloo Region that advances a new model of attainable and affordable housing. Through interdisciplinary research and training, collaborative "Quadruple Helix" partnerships, and data-driven urban decision-making, the Living Lab will assess and amplify the economic, health, sustainability, technological and societal impacts of community-led housing — creating local solutions with global relevance.

LIVING LABS: Living Labs are real-world research and innovation environments where end-users are active participants in co-creating and testing solutions to the world's most pressing issues. Living Labs foster innovation among main actors in the Quadruple Helix Model: citizens, government, academia, and industry. As a core principle, our Living Lab is committed to open data, knowledge mobilization, experiential learning opportunities for students, authentic community partnerships and democratization of the tools and recommendations we generate.

CONTEXT: Build Now Waterloo Region is a community-led housing initiative that plans to build 10,000 new "missing middle" homes in Waterloo Region, Ontario by 2030 (70% ownership, 30% rental). Home ownership has been one of the most important wealth building and economic prosperity engines for families in North America for decades. However, in the past decade, home ownership has become out of reach for many young Canadians, and represents a national security threat, according to the RCMP. However, to date, attempts to create affordable home *ownership* opportunities have been limited to small developments led by Habitat for Humanity. Affordable housing projects in Canada are overwhelmingly focused on rentals, which do not confer the same economic or health and well-being related benefits as home ownership. Build Now is led by Habitat for Humanity but represents a strong coalition between two local developers, a local construction association, the municipalities, a local housing cooperative, and the Future Cities Institute. To date, the team has two parcels of land entrusted by Kitchener and Waterloo and they are actively seeking other municipal partnerships for additional parcels. The first parcel on which they will build over 1000 units of housing is a 25-acre/$20 million parcel in north Waterloo, with the first 50 units needing to be approved and permitted by December 2025. They have not yet broken ground.

No urban problem is unique. Problems facing the Region of Waterloo are facing Canadian communities from Vancouver to St. John's, and from Windsor to Iqaluit. Although we are using Build Now as a case study in this proposal, findings will have relevance to all cities in Canada. The FCI is deeply committed to knowledge mobilization, and will create a repository of open access reports, tools, and recommendations accessible for all Canadian communities.

ALIGNMENT WITH ENVIRONMENT 2035:

1. **How does the initiative outlined in your LOI contribute to Environment 2035 and/or Environment's role in Global Futures Networks?**

By co-creating affordable housing solutions with Habitat for Humanity, we aim to:

- Accelerate innovation in urban sustainability and climate adaptation
- Strengthen research connections both within ENV and with other faculties
- Train the next generation of interdisciplinary leaders
- Showcase solutions that are scalable across municipalities
- Foster strong, transparent partnerships between academia, industry, and community

In terms of alignment with Environment 2035, the following are examples of how this project aligns with each of the four priority areas.

1. Climate Change Adaptation and Mitigation
    - Heat and flood risk mitigation
    - Climate-resilient design and natural infrastructure
    - Renewable energy integration
2. Cities of the Future
    - Housing innovation
    - Livability and inclusive urban design
    - Smart, sustainable infrastructure and digital planning
3. Natural Resources and Environment
    - Urban biodiversity and water management
    - Green and blue infrastructure in residential developments
    - Land use for resilience and sustainability
4. Sustainable Transitions: Energy and Finance
    - Green infrastructure financing models
    - Business models and economic assessment for municipal sustainability investments
    - Distributed energy load interventions in residential settings

2. **What resources/supports (if any) do you anticipate needing from the Faculty of Environment should your project move ahead?**

None beyond what the Faculty is already doing to support the Future Cities Institute.

INNOVATION: Build Now represents a $500 million investment in our Region that is moving quickly. Table 1 describes housing innovations unique to the Build Now initiative within the realm of affordable housing projects in North America.

Table 1. Build Now innovation compared to standard practice for affordable housing projects

	Standard Practice	Build Now
Affordable housing focus	Rentals	Predominantly ownership (70% of units)
Initiative partners	Governments, social enterprises or not-for-profits	Developers, governments, social enterprises and not-for-profits, construction association, and academia
Trust	Lack of trust between municipalities/not-for-profits and for-profit developers	Building trust between all partners has been a key component of the work to date
Procurement	Municipality-led affordable housing projects require procurement practices that result in a cost of development for well above market prices.	Opening the procurement pipeline through partnership with the Grand Valley Construction Association means that cost of development can be procured for fair market prices.

Figure 1 shows the conceptual site plan presented to the City of Waterloo for the 25 acre parcel. Buildings range from stacked townhomes to 12 storey buildings, with greenspace and "plaza" or "public square" space prioritized.

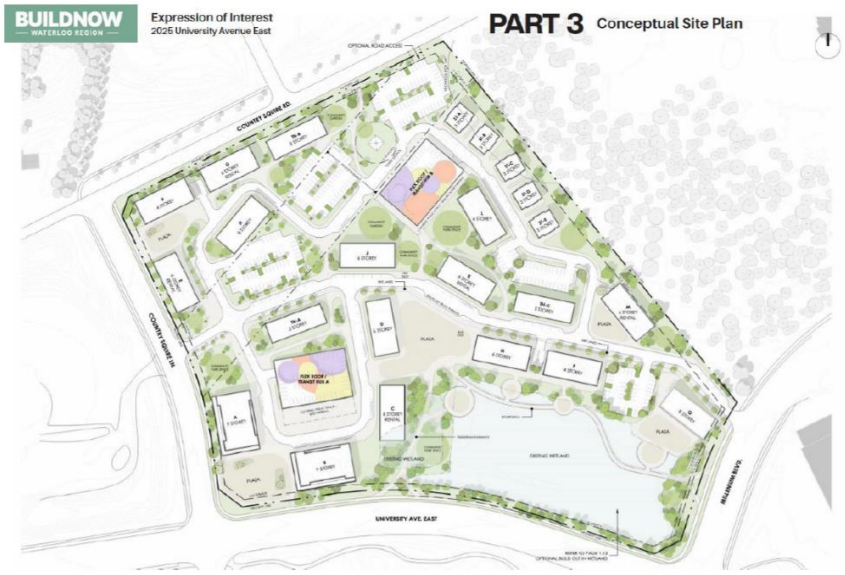

Figure 1. Conceptual Site Plan for Build Now 25-acre build (1010 units at 2025 University Ave. E)

BUDGET

Global Futures Funding will leverage over $500,000 in external funding. The team is also submitting an NFRF Transformation proposal in the amount of approximately $20 million, and will be actively pursuing NSERC, CIHR and SSHRC funding over the next five years, as well as other governmental and industry sponsored research aligned with the project.

Table 2: Budget

CATEGORY	TOTAL PROJECT BUDGET	GFF Request Year 1 (2024-2025)	GFF Request Year 2 (2025-2026)	GFF Request Year 3 (2026-2027)	FUNDED BY OTHERS
RESEARCH					
GFN Coordinator	$275,436.00	$90,000.00	$91,800.00	$93,636.00	$0.00
Global Futures Seed Funds	$990,000.00	$90,000.00	$90,000.00	$90,000.00	$60,000.00
TRAINING					
Interdisciplinary Capstone PLUS Projects	$150,000.00	$25,000.00	$25,000.00	$25,000.00	$75,000.00
CIVIC	$302,400.00	$10,800.00	$10,800.00	$10,800.00	$270,000.00
Build Now Innovation Challenge	$126,000.00	$25,000.00	$25,000.00	$25,000.00	$51,000.00
Build Now Problem Lab	$30,000.00	$0.00	$0.00	$0.00	$30,000.00
SERVICES AND PARTNERSHIP DEVELOPMENT					
City Sprints	$45,000.00	$5,000.00	$5,000.00	$5,000.00	$30,000.00
European Network of Living Labs (ENoLL) In-Person Supports	$6,600.00	$4,200.00	$2,400.00		
TOTAL	**$1,265,436.00**	**$250,000.00**	**$250,000.00**	**$249,436.00**	**$516,000.00**

GFN Coordinator (Coordinator). The Coordinator, working at the FCI and reporting to the Managing Director, provides overall coordination to the research, training and service activities described below. The Coordinator will be a USG 8, and would be a net new staff member to the FCI (contingent on funding). The Coordinator will have experience in coordinating large interdisciplinary research teams with large budgets. (~$90,000/year)

Research

Global Futures Seed Funds. FCI will coordinate seed funding for interdisciplinary project ideas aligned with Build Now activities. To obtain funding, research projects must be based on questions that have been co-created with Build Now and/or other external community partners. Each research team must include at least two faculty members from different faculties, and must include a Build Now team member or community member. Eligible proposals will include applied research that directly addresses issues of importance to external partners.

We are allocating $15,000 per year for each of the five global futures plus the Emerging Opportunities Team. ($90,000/year)

Training

Interdisciplinary Capstone Plus (i-Cap+). The FCI helped coordinate the University's first fully interdisciplinary capstones that included approximately 120 students from all six faculties, with the theme of Future Cities. The FCI leveraged its relationships with municipalities, industry and NGO partners to elicit project ideas, which were presented to students. The Provost's Interdisciplinary Funding was awarded to Dean Wells for 2024/2025 and 2025/2026 to establish and set up processes for the i-Cap. $37,500 per year of the budget was allocated to the FCI, which paid for existing staff time to a) engage in business development with municipal, industry and civil-society partners to submit project ideas; b) organize the launch and the final i-Cap event; c) coordinate meetings between external partners and student groups.

In 2025/2026 and beyond, the FCI will build on the i-Cap success by undertaking the i-Cap PLUS. Briefly, the i-Cap PLUS will coordinate Future Cities-related projects both for i-Cap students but also for other capstone projects around the campus. For example, in 2025, fourth year Urban Planning students had their standard 4-month capstone project, did not participate in the Future Cities i-Cap. The FCI will seek to engage other capstone instructors and students across the Faculties to engage in projects related to Build Now and/or other future cities-related projects. ($25,000 per year)

CIVIC Response Team. FCI recently launched the CIVIC initiative: City/Community Innovation via Internships & Co-op. Industry partners and municipalities fund CIVIC positions, and financial resources are matched via MITACS and other sources to maximize value of external partner contributions. This collaborative framework benefits both the partner and the broader research community. Between September 2024 and September 2025, the FCI will have hired 9 co-op students to support research projects driven by external partners. As we continue to build our networks with municipalities, it is becoming clearer that solid partnerships rely on our timely responses to city needs. Therefore, the FCI is investing in a continuous stream of talented co-op students who can push projects forward quickly.

The FCI will hire two senior-year co-op students per term to respond to research and training needs identified by both internal and external partners. The co-op students each term will be from different faculties, but they will work together on emerging issues to increase their interdisciplinary training experience at Waterloo. The FCI will leverage external partner funding (at $45,000/year), MITACS funding (at $45,000 per year) and Global Futures Funds (at $11,000 per year) to cover the cost of six co-op students per year. ($11,000 per year).

Build Now Innovation Challenge. Over the course of two weeks, University of Waterloo students will team up across disciplines to take on real-world challenges through the Build Now Innovation Challenge. Focused on accelerating local housing solutions, this challenge invites students to develop bold, creative ideas that tackle some of the most urgent issues in housing and community development. The Innovation Challenge creates a fast-paced, low-risk environment where students can apply classroom learning and cutting-edge research to tangible, place-based problems. By working closely with municipal staff, housing experts, and community partners, students receive rapid feedback and valuable insights, learning how to turn their ideas into practical, scalable solutions. ($25,000 per year)

Future Builders Tour: Exploring Circularity, Community & Construction. The *Future Builders Tour* is an immersive, multi-stop experience designed to introduce students to real-world affordable housing projects across Waterloo Region. Open to students from all faculties with an interest in impact-driven community housing innovation, this annual experience connects participants with three key sites:

- **Habitat for Humanity ReStore** (Waterloo) – Learn how circular economy principles are applied through reclaimed materials, resale strategies, and community programming.

- **Habitat for Humanity Build Sites** (Cambridge/Kitchener/Waterloo) – Visit active construction sites and see how accessible housing is being built with volunteers, students, and sustainable materials.
- **BUILD NOW Innovation Site** (Waterloo) – Explore the frontlines of modular and net-zero housing innovation. Learn how rapid, scalable solutions are being developed to address Canada's housing crisis.

This tour will be first piloted in Summer 2025 in partnership with the Engineering Ideas Clinic's Sustainability Fellowship, including participation from first year We Accelerate students, international capstone students, and upper-year engineering students. After the initial pilot, we would like to expand access and offer the *Future Builders Tour* to a broader range of students through a recurring annual program.

Build Now Problem Lab. At the intersection of Training and Partnership Development, the Problem Lab at the University of Waterloo will host two Build Now Problem Labs per year. Students, external partners, and faculty will engage in three-hour workshops to more deeply understand a particular problem related to affordable housing innovation. The Problem Lab will provide in-kind contributions for the first Build Now problem lab ($5000 in-kind contribution), with our team raising external funds of $5000 per Problem Lab thereafter.

Service and Partnership Development

City Sprints. City Sprints are an FCI initiative, adapted from Google Ventures' Design Sprint Methods, that bring together municipal, industry, and civil society partners to focus on problem solving over a 1-2 day time frame. Participants are given reading materials and problem statements in advance of the Sprint, and move through activities from Mapping the Problem to Sketching Solutions, Storyboarding Solutions and Next Steps. The FCI will co-host the meetings with the Waterloo Region Community Foundation, which will contribute $10,000/year towards the organization and creation of work packages that result from these workshops. ($5000 per year)

ENoLL Capacity Development. To set up our Living Lab, we will leverage the supports provided by the European Network of Living Labs (ENoLL), a global leader in fostering user-centered, real-world innovation environments. We plan to participate in ENoLL's capacity-building programs, including the Virtual Learning Lab and their foundational toolkits like the Living Lab Handbook, to strengthen our team's capabilities in co-creation and applied research. Through ENoLL's global network, we will access a vibrant community of Living Labs, enabling knowledge exchange and potential collaboration on international projects. ENoLL will guide our development from concept to implementation, ensuring our Living Lab is both impactful and future-focused.

Organization: This project will be led by five Global Futures Teams led by faculty from across campus, plus one "Emerging Opportunities" Team. Given that housing is a critical and urgent priority in Canada, the Emerging Opportunities Team will be responsible for identifying and supporting pilot research and partnership development with municipalities and organizations aligned with Build Now objectives.

University Constituents/Partners

Faculty members from all six Faculties are involved, as are both research stream faculty, teaching stream faculty, and Associate Deans work-integrated learning. External to the university, our local partners include government, industry, and civil society organizations. Partners are described in Table 3.

Table 3. University team leads

Role	Faculty	Name
University of Waterloo – Research Leads		
Lead – Health Futures Cluster	Health	Troy Glover
Lead – Technological Futures Cluster	Math	David Del Rey Fernandez
Lead – Sustainability Futures Cluster	Environment	Michael Drescher
Lead – Economic Futures Cluster	Arts	Anindya Sen
Lead – Societal Futures	Environment	Katherine Perrott
Lead – Emerging Opportunities Team	Environment	Leia Minaker
University of Waterloo – Training Leads		
Co-Lead – Training	Engineering	Derek Rayside
Co-Lead – Training	Environment	Michael Wood

IMPACT: Natural experiments in city building are rare and precious opportunities. Unlike simply observing change in one location over time, a longitudinal study with multiple interventions (or land parcel developments) in diverse locations will allow our team to use a rigorous study design (that allows us to approach causal attributions) to create world-class data on how health, economic outcomes, societal goods and policy, technology, and sustainability change over time as more homes get built.

Habitat for Humanity has 50 active affiliates across Canada, serving almost 400 communities. This is a once-in-a-generation opportunity to come in at the ground floor and measure everything that happens – relevant to all Global Futures – over time. Being the academic partner on this first-of-its-kind home-building partnership between civil society, government, and industry before the first shovel has broken ground allows for an unprecedented, huge, collaborative research undertaking that will be interdisciplinary, efficient (researchers can leverage the project to apply for additional funding), and globally

relevant. FCI's current partnerships with the Horizon Europe team and African cities mean that we can share learnings and learn from global partners.

Global Futures Alignment

Aligned with Waterloo at 100, Waterloo has identified five Global Futures: Technological futures, Economic futures, Societal futures, Health futures, and Sustainable futures. Each of the Global Futures intersects with the others in complex ways. Table 1 shows how the Build Now Living Labs either directly or indirectly align with the Global Futures.

Table 1. Direct and indirect alignment Build Now Living Lab Network and Global Futures

Global Future	Build Now LL
Technological Futures	
Direct	Tech innovations born and tested at Waterloo LL can be scaled up in Build Now communities
Indirect	Social acceptance of tech innovations tested over 10 years
Economic Futures	
Direct	Cost and benefits of Build Now developments are rigorously evaluated
Indirect	Impact of Build Now on Regional housing affordability; 10 year longitudinal study of residents identifies multiple household economic outcomes
Societal Futures	
Direct	Build Now negotiations with municipalities and trust building are carefully documented, including changes to policy and new regulations
Indirect	10 year longitudinal study of residents identifies multiple societal outcomes associated with Build Now developments
Health Futures	
Direct	10 year longitudinal study of residents identifies multiple health outcomes associated with Build Now development features (e.g., urban design, mixed land use, mobility)
Indirect	Assessment of healthy built environment and transportation options
Sustainable Futures	
Direct	Prototyping and testing community energy use and mobility interventions, sustainable infrastructure, green and blue infrastructure in residential developments.
Indirect	How are sustainability-related behaviors of residents tied to other pro-social behaviors?

Appendix A

Global Futures | Waterloo at 100

	Health	Economic	Societal	Technological	Sustainable
Phase 1	• Neighbourhood-level measures	• Neighbourhood-level measures	• Neighbourhood-level measures • Planning policy assessment and changes	• Neighbourhood-level measures	• Sensors; land images; soil quality? Water quality? Air quality? (other stuff to ask Chris or others about) • Biodiversity
Phase 2	• Noise pollution	• Jobs created through construction • Construction worker safety		• New technology to optimize concrete usage • Pipe-network sensors	• Sustainability outcomes related to housing construction • Water quality treatment (?) • RISE scorecard? • Infrastructure
Phase 3	• Baseline surveys of residents with residential history; health history, etc.			• Sensors to monitor all other outcomes (e.g., traffic,	• Ecological resilience • Assess built/natural environments
Phase 4	• Ongoing residential surveys • Aging in place • Child health • Mental health and addictions – System transformation work	• Return on investment • Quantifying and monetizing health outcomes (Larry Frank) • Local GDP • Business starts or entrepreneurship • Insurance (Jason T) • Housing market changes in cities	• Fertility • Children's educational attainment • Political polarization • Social inclusion • Residential satisfaction • Planning policy changes • Culture and entertainment • Crime rates	• Using technology and data for data-driven decision making. • Linking and modeling complex data • Effective feedback loop to feed information back to city	• Transportation modeling • Biodiversity • Change in energy, water use over time